

Mid-Project Review — Speaker Notes

Per-Class Threshold Calibration for Training-Free Open-Vocabulary Semantic Segmentation of Remote Sensing Imagery

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Companion to `review.pdf`

How to use this. One entry per slide, in order. Each gives the purpose of the slide, the line of argument to deliver, and the question most likely to follow. Total speaking time at this level of detail is roughly **18–22 minutes**, which leaves room for discussion in a thirty-minute slot.

The single thread running through the talk: the pipeline is not wrong about *what* it sees; it is wrong about *when to commit*. Every slide either establishes that, quantifies it, or corrects it.

1. Title

What to say. Give the title, the two of you, and the supervisors. One sentence of orientation: “This work improves an existing open-vocabulary segmentation pipeline for aerial imagery without training any network weights.” Then move on — do not explain the method here.

2. Presentation Outline

What to say. Read the eight section names in about ten seconds. The purpose is to signal that the mandated structure is followed, not to summarise.

3. Introduction

Why this slide exists. The committee may not know this sub-field. Three things must land before anything else: what the task is, what the baseline does, and what we changed.

What to say. “In ordinary semantic segmentation the class list is fixed at training time. In *open-vocabulary* segmentation the class list is an *input* — you supply words at inference, and the model segments them. No weights are trained; the vocabulary can change from one image to the next.”

“Our baseline is SegEarth-OV3, which applies Meta’s SAM 3 to remote sensing. For each word you give it, SAM 3 returns a score map. The pipeline then takes the highest-scoring class per pixel, and if that score is below a threshold τ it discards the answer and calls the pixel ‘background’.”

“We reproduced their published result exactly — **47.38 against a published 47.4** — so every number after this is measured against a baseline we control, on our own hardware.”

“Our contribution is to that final decision: one threshold per class, and one scale per class before the comparison. **No weights are trained.**”

Likely question: *Why not just train the model?*

Answer: Because the setting is training-free by definition: the class list is not known in advance, so there is nothing fixed to train for. Fine-tuning also needs labelled data in the thousands of tiles, which is exactly what a new region does not have.

Be careful. Do not say “unsupervised”. The vocabulary is given, so the correct terms are *training-free* and *annotation-free*.

4. Motivation — the baseline discards land cover

Why this slide exists. This is the slide that justifies the entire project. If the committee accepts that a large, recoverable residual exists and that no simple fix reaches it, everything downstream follows.

What to say. “We instrumented the baseline to ask a question its own paper does not: how much real land cover does it throw away? On LoveDA, at their own published threshold, **29.68% of all real-class pixels** — 323 million — are assigned to the catch-all class.”

“More precisely: the pipeline is not mostly *wrong*, it is mostly *silent*. Discard exceeds genuine class confusion by three to one.”

“The clearest case is **water**. Precision **89.5%**, recall **54.7%**. When it says water it is right nine times out of ten — and it finds barely half the water in the image. Those missing pixels are not misclassified. They fall below the threshold and are discarded.”

“The obvious response is to lower the threshold. We tried it: **−5.54 mIoU**, because you gain 1.73 wrong pixels for every right one. We also tried removing the presence gate: **−11.97**. Both global adjustments cost more than they return.”

“So the correction cannot be global. **It has to act per class** — which is the rest of the talk.”

Likely question: *Isn't a low recall just the model being bad at water?*

Answer: No, and the precision number is why. A model that is genuinely bad at a class is wrong when it fires. This one is right 89.5% of the time it fires. The information is present; the decision rule refuses to use it.

5. Motivation — the discard, visualised

Why this slide exists. The previous slide is numbers. This is the same fact as a picture, and it is what people remember.

What to say. “Four panels, one LoveDA tile. Input, ground truth, the baseline’s prediction, and in the last panel the pixels it sent to background.”

“Look at the ground truth: the top of the tile is water. Look at the prediction: that region is gone. And in the last panel you can see it was *discarded*, not confused with another class — **19% of this single tile**.”

“The threshold that caused this was tuned for the dataset as a whole. It is a reasonable value on average and the wrong value for water.”

Likely question: *Is this tile representative?*

Answer: It is one tile chosen to show the mechanism clearly. The dataset-wide figure is the 29.68% on the previous slide, measured over all 1669 validation images.

6. Application domains

Why this slide exists. Establishes that the training-free setting is not an academic convenience but the operational reality in several domains.

What to say. “Where does an open vocabulary actually matter? Three cases.”

“**Disaster response.** After a flood you have UAV imagery within hours and no labelled data for that district. You cannot train anything in time.”

“**Infrastructure.** A class like **solar panel** appears in no land-cover benchmark. Here you can simply type it.”

“**Settlement mapping.** The vocabulary changes between surveys — **tin roof, unpaved road, water tank**.”

“The relevant number is the deployment cost: our correction needs **about 200 labelled tiles and no training run**. Fine-tuning instead would need thousands of labelled tiles and would forfeit the training-free property that made the pipeline attractive.”

7. Problem statement — formulation

Why this slide exists. The committee asked for a mathematical formulation. This slide is that, and it should take under two minutes.

What to say. “ $s_c(p)$ is the score class c gets at pixel p . The baseline takes the **argmax** — the highest-scoring class — and keeps it only if that score clears one global τ ; otherwise the pixel becomes the catch-all class b .”

“**argmax** simply means ‘which class scored highest’. **max** gives the value, **argmax** gives which one it was.”

“We change two things. First, τ becomes a *vector*: each class gets its own threshold. Second, we multiply each class’s score by a fitted weight w_c *before* the comparison, so the weight can change *which* class wins.”

“Both are fitted by coordinate ascent on about 200 labelled tiles, and — this is the point — **no network weights are modified**. The threshold is applied to the raw score, so w only affects the **argmax**.”

Likely question: *Isn’t fitting on labelled data just supervision through the back door?*

Answer: The baseline already tunes its single τ on labelled data — 0.5 for LoveDA, 0.1 for OpenEarthMap. We spend the same kind of budget on N parameters instead of one. It is the same protocol, not a new one.

8. Completeness of the per-class threshold family

Why this slide exists. **This is the intellectual core of the work.** If a committee member asks “what is actually novel here?”, this slide is the answer. Do not rush it.

What to say. “A natural objection is: why thresholds? Why not recalibrate the scores in some cleverer way?”

“The proposition answers that. Take *any* strictly increasing per-class transformation of the scores, and compare against a global threshold. Because the transformation is increasing, you can invert it — and the comparison becomes exactly a per-class threshold.”

“So **per-class thresholds are not one option among many. They are the complete family** of everything reachable by rescaling scores *after* the **argmax**. Nothing in that family can beat them.”

“Scaling *before* the **argmax** is different, because it changes which class wins. That is a strictly larger family — and it is not decoration. We measured that **6% of discarded pixels are cases where the catch-all class won the argmax outright**. A threshold cannot change an **argmax**, so no threshold anywhere can fix those. That is precisely what the second parameter set is for.”

Likely question: *So the second lever subsumes the first — why keep both?*

Answer: They are fitted together and reported as two stages so the contribution of each is visible. The first stage is also the one with the completeness guarantee, which is worth stating separately.

9. Literature survey

Why this slide exists. Committee requirement: a minimum of ten works. Fourteen are listed.

What to say. Do not read the table. Walk the three groups in about a minute:

- **Rows 1–7** — how dense prediction was extracted from vision–language models: CLIP, then MaskCLIP removing pooling, then a line of attention corrections.

- **Rows 8–11** — the remote-sensing line, ending at our baseline SegEarth-OV3, and ConInfer as the nearest contemporary work.
- **Rows 12–14** — the tools we borrow: calibration, classical thresholding, classical spatial refinement.

Be careful. You have read SegEarth-OV3, ConInfer and SAM 3 in full. If asked about a row you have only skimmed, say so plainly and describe what it contributes — that is a normal and acceptable answer.

10. Research gaps

Why this slide exists. Connects the literature to the contribution. Each gap should map to something the work then does.

What to say. “**Gap one.** Every method in this family ends the same way: an `argmax`, then one global threshold. All the research effort goes into producing better *scores*. Nobody has asked whether the *rule* applied to those scores is the right shape.”

“**Gap two.** Papers report mIoU. None reports how much real land cover ends up in the catch-all class, or whether it can be recovered. We measured it.”

“**Gap three.** mIoU averages classes *equally*. So a ‘background’ class — which in LoveDA means ‘everything else’ — owns one seventh of the score regardless of whether it means anything. A method can appear to improve or degrade while land-cover quality is unchanged. We observed this in both directions and therefore report two metrics throughout.”

11. Objectives

Why this slide exists. Separates what is finished from what is not. **Do not present the project as complete.**

What to say. “Three objectives are met. Reproduce the baseline exactly, so our numbers mean something. Quantify what it discards and test whether any global adjustment recovers it — none does. And design a correction that trains no weights.”

“Three remain open. The fitted parameters are currently *constant across tiles*, so they are right on average and can be wrong on an individual scene; a scene-adaptive version is the natural next step. SAM 3’s two heads are combined by a fixed maximum, which is another unexamined decision of the same kind. And the evaluation needs widening to establish when the correction transfers.”

Likely question: *What is the biggest remaining risk?*

Answer: Transfer. The fitted parameters are specific to a data distribution, so a deployment must calibrate on the distribution it will run on. Characterising that boundary properly is the main open item.

12. Experimental setup

Why this slide exists. Establishes that the evaluation is complete and honest before any result is shown.

What to say. “Three datasets, and **every image in each is used**. LoveDA validation is 1669 tiles at 0.3 m; Potsdam is 2016 tiles at 5 cm; OpenEarthMap is 384 tiles.”

“Two points about the protocol matter. **Calibration and evaluation tiles are always disjoint** — we never fit and test on the same image. And the baseline is *recomputed* on the same held-out tiles, so the two rules are compared on identical pixels rather than against a published number.”

“We also keep a validation gate: any instrumented run must still reproduce 47.37 mIoU and 29.68% discard. If either moves, the instrumentation changed the model and the run is discarded.”

Likely question: *Why only 384 tiles of OpenEarthMap?*

Answer: That is the public redistribution we could obtain; the official split is 500. We say so wherever that dataset appears.

13. Results

Why this slide exists. The main quantitative claim.

What to say. “Read the table by column. On LoveDA the two stages give **+1.18** and **+1.16**, total **+2.32**. On Potsdam, **+5.67**. On OpenEarthMap the headline is +0.30 — but under the catch-all-excluded metric it is **+1.75**, larger than LoveDA’s. That is gap three from earlier, appearing in our own results.”

“**Every fold is positive** on both datasets where we ran the full protocol.”

“The row that matters most is the verification. These gains were first computed from a cached statistic. We then ran the actual pipeline end to end on Potsdam: **57.60** → **58.35** → **63.27** on 1816 held-out tiles, **every stage within 0.05 mIoU** of what we predicted.”

“The largest single change is Potsdam **tree**: **37.92 to 59.09**. Its precision is 93.34% and its recall 38.63% — the same asymmetry as **water** on LoveDA, and exactly what the method is built to correct.”

Likely question: *Is a one-to-two point gain significant?*

Answer: Five-fold cross-validation with every fold positive, and the Potsdam gain is +5.67. For comparison, the gap between the baseline and the previous state of the art in this family is of the same order — and this costs no training.

14. Results — qualitative

Why this slide exists. Shows the correction acting, and shows a failure. **Point at the failure before anyone else does.**

What to say. “Four Potsdam tiles. Input, ground truth, baseline, ours. **The network, its weights and the forward pass are identical between the last two columns** — only the decision rule differs.”

“Row one: a normal urban scene. **tree** improves, and the classes that were already well calibrated — **building**, **car** — are left within a point of where they were. The correction does not churn.”

“Row two: the baseline misses a canopy entirely; the correction recovers it.”

“Row three: fourteen pixels change out of a quarter of a million. Where nothing is miscalibrated, nothing happens.”

“**Row four is a tile where we lose 7.72 mIoU**. It is in the figure deliberately. Our parameters are constant across tiles, so they are correct on average — +5.67 over the dataset — and can be wrong on one scene. That is the limitation, and it is the open objective on the previous slide.”

15. Application study

Why this slide exists. Demonstrates the method outside the benchmarks, on imagery from India.

What to say. “25 tiles from 5 openly licensed UAV scenes over India at 4–5 cm — a region in none of the three benchmarks. We ran it unchanged under two different class lists, typed at inference.”

“The finding is the *stability*. A flooded building is partitioned almost identically by both vocabularies: 78.6% under **water** against 79.1% under **flood water**; 18.2% under **concrete roof** against 18.3% under **building**. **The same regions are recovered; only the names follow the words supplied.**”

“The second case is a failure and worth stating. A solar farm: with no word for panels, 100.0% of the tile goes to one class. Supply `solar panel` and it recovers 4.6% — *below the panels’ visible extent*.”

“There is no ground truth for this imagery, so these are pixel shares, not accuracy. We make no accuracy claim on these scenes.”

Likely question: *Why not calibrate on the Indian imagery too?*

Answer: It is unlabelled, and our transfer results show that parameters fitted elsewhere would not help. The 200 labelled tiles are what such a deployment would have to budget for — which is the honest cost of the method.

16. Application study — one scene, two vocabularies

What to say. “Same weights, same forward pass. Only the class list differs between the second and third columns. Lower row, the flooded building — recovered consistently under both. Upper row, the photovoltaic array, for which the first vocabulary has no corresponding class at all.”

17. Project timeline

What to say. “Work to date: the baseline and the measurement in August; the two parameter sets and their verification through September; the application study this month. Also in August we built a semantic co-occurrence prior, evaluated it and *rejected* it — it did not earn its place, and that is recorded.”

“Remaining: the scene-adaptive formulation and the head-fusion question through December, then writing, with submission targeted at a journal in the first quarter of 2027.”

Likely question: *Why was the co-occurrence prior dropped?*

Answer: We measured it against a plain neighbour vote. It added two tenths of a point, and a *perfect* matrix added three tenths. The idea was sound and the measurement did not support it, so it is reported as an ablation rather than a contribution.

18. References and closing

What to say. Do not read them. “Fourteen key references; the full bibliography is in the manuscript.” Then thank the committee and invite questions.

Three things to have ready

1. **“What is novel?”** — the completeness proposition. Per-class thresholds are the complete family given a fixed `argmax`; scaling before the `argmax` is provably larger, and reaches the 6% of errors a threshold cannot.
2. **“Is this just a SAM 3 quirk?”** — no. The same correction was applied to a CLIP-based pipeline, where it is worth $+2.51 \pm 0.34$ mIoU, larger than on SAM 3. One architecture is a result; two is a property.
3. **“What is left?”** — scene-adaptive parameters, the head-fusion question, and characterising when the correction transfers. Say this plainly; the work is not finished.